

EES102: Introduction to Environmental Sciences

Managing our waste

Wastewater treatment, Solid waste management, Waste as a Resource

by

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Managing our waste

- Liquid waste/wastewater treatment
- Solid waste management
- Resource Recovery from wastes

Relates to **almost all Sustainable Development Goals (SDGs) directly or indirectly**, but particularly relevant to [#11](#) (Sustainable Cities and Communities), [#12](#) (Responsible Consumption and Production), and [#6](#) (Clean Water and Sanitation)

Environmental sustainability

Consumption of resources	State of the environment	Sustainability
> nature's ability to replenish	Environmental degradation	Not sustainable
= to nature's ability to replenish	Environmental equilibrium	Steady state
< nature's ability to replenish	Environmental renewal	Sustainable

- Transitioning from a linear model of resource utilization to a circular one

Broader 3R concept in waste management

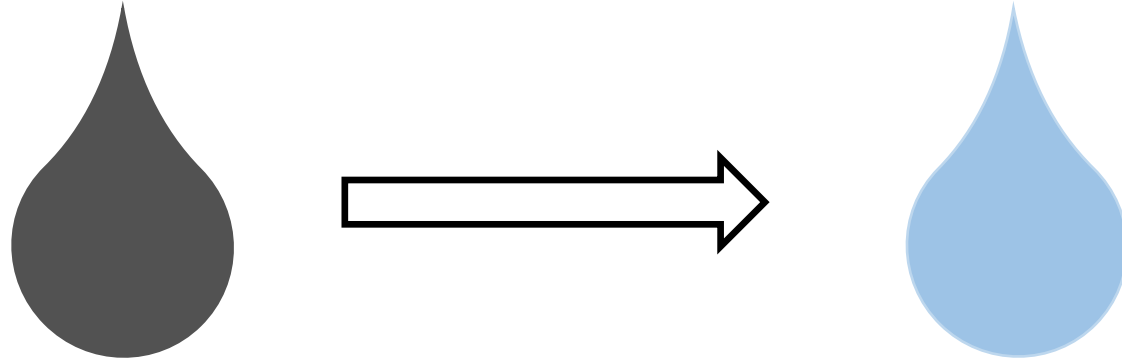
- Reduce
- Reuse
- Recycle

to ensure sustainable and effective
use of resources and materials

Poster by
Harpreet Singh
(MS20029)



Wastewater treatment



watertechonline.com



www.unwater.org

What is wastewater?

- Water that emerges after freshwater use by humans for domestic, commercial and industrial purposes

OR

Water that is no longer suitable for its most recent use due to the presence of various undesirable components

- Municipal wastewater/Domestic wastewater/Sewage
- Industrial/Commercial process wastewater (usually requires a specialized treatment process)

Why should wastewater be handled/managed/treated?

- To avoid **lowering of dissolved oxygen** (e.g., for aquatic life, DO should be at least 4-5 mg/L)
- Avoid adding nutrients (N and P) to water bodies that cause excessive growth of photosynthetic microbes → **algal blooms** → **anaerobic zones** → **dead zones**
- Avoid increasing the **load of suspended solids** or sediments in streams (turbidity increase affects aquatic life)
- To lower **odor in the stream** (caused mainly by anaerobic activities)
- Avoid **contamination of freshwater reservoirs** and associated health hazards
- Water **conservation and reuse**

Key constituents of domestic wastewater

- Total solids (suspended + dissolved)

- Organic matter

- Inorganic matter

- Microorganisms

- Other components

(e.g., pharmaceutical residues,
surfactants, oil and grease, paints,
emerging contaminants, etc.)

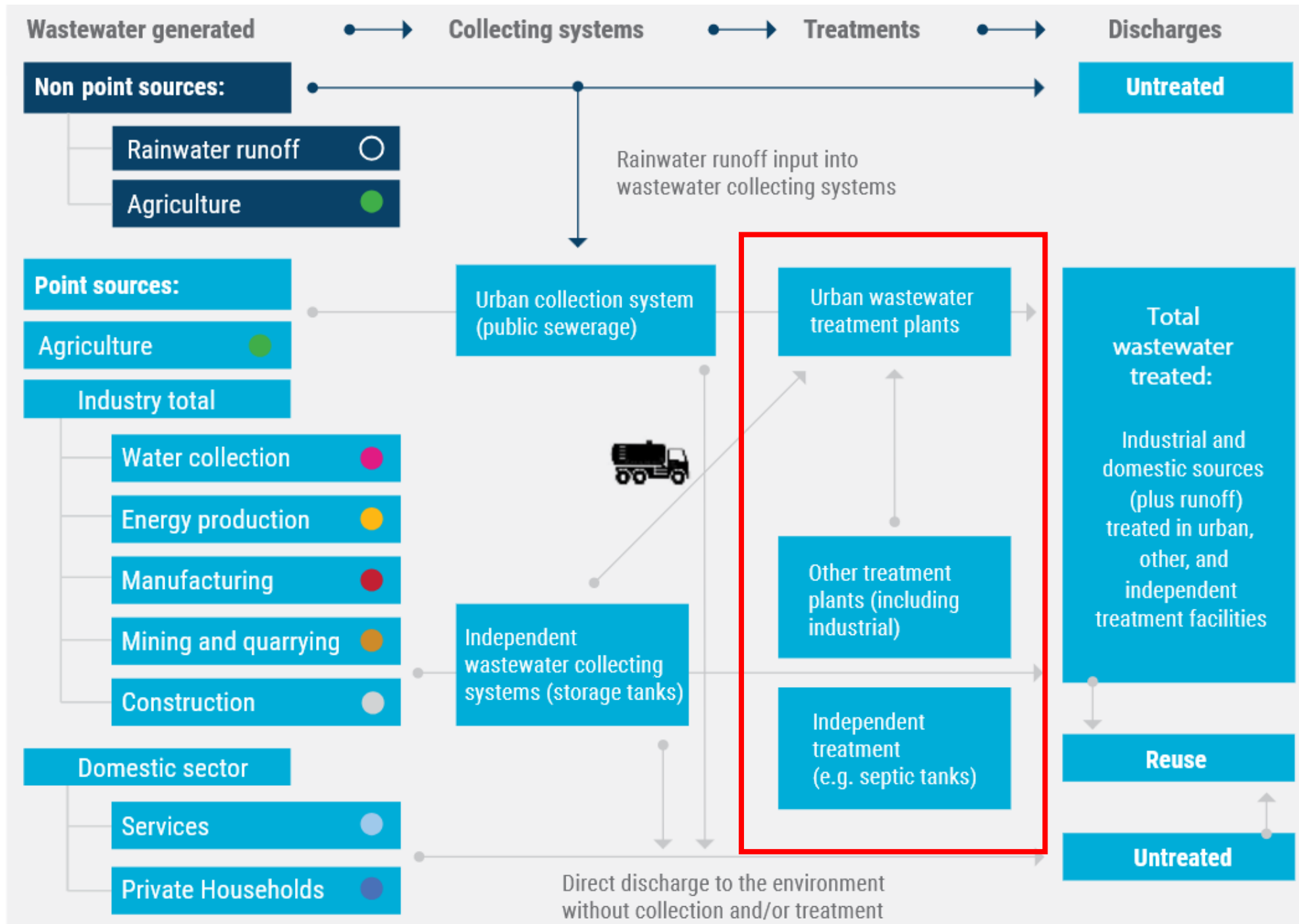
- Chemical oxygen demand (COD)
- Biochemical oxygen demand (BOD)
- BOD:COD ratio

- Nutrients like P and N
- Heavy metals

Typical domestic wastewater composition and strength

Constituent <i>Strength</i>	Concentration (mg/L)		
	<i>Low</i>	<i>Medium</i>	<i>High</i>
Total solids	350	700	1200
Total dissolved solids (TDS)	250	500	850
Total suspended solids (TSS)	100	200	350
COD	250	500	1000
BOD ₅	110	220	400
Total Nitrogen (TN)	20	40	85
Total Phosphorous (TP)	6	10	20
Chlorides	30	50	100
Alkalinity (as CaCO ₃)	50	100	200
Oil and grease	50	100	150
Sulphate	20	30	50
Total coliforms (MPN/100 mL)	10 ⁶ -10 ⁷	10 ⁷ -10 ⁸	10 ⁸ -10 ⁹

Flow diagram

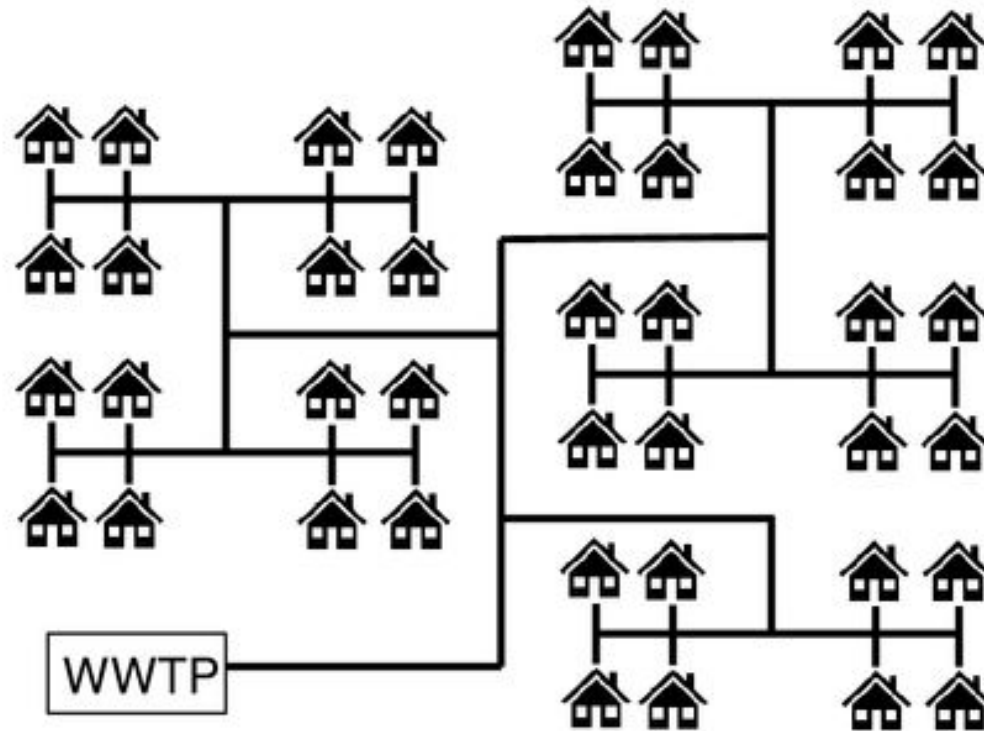


Source:
UN Water

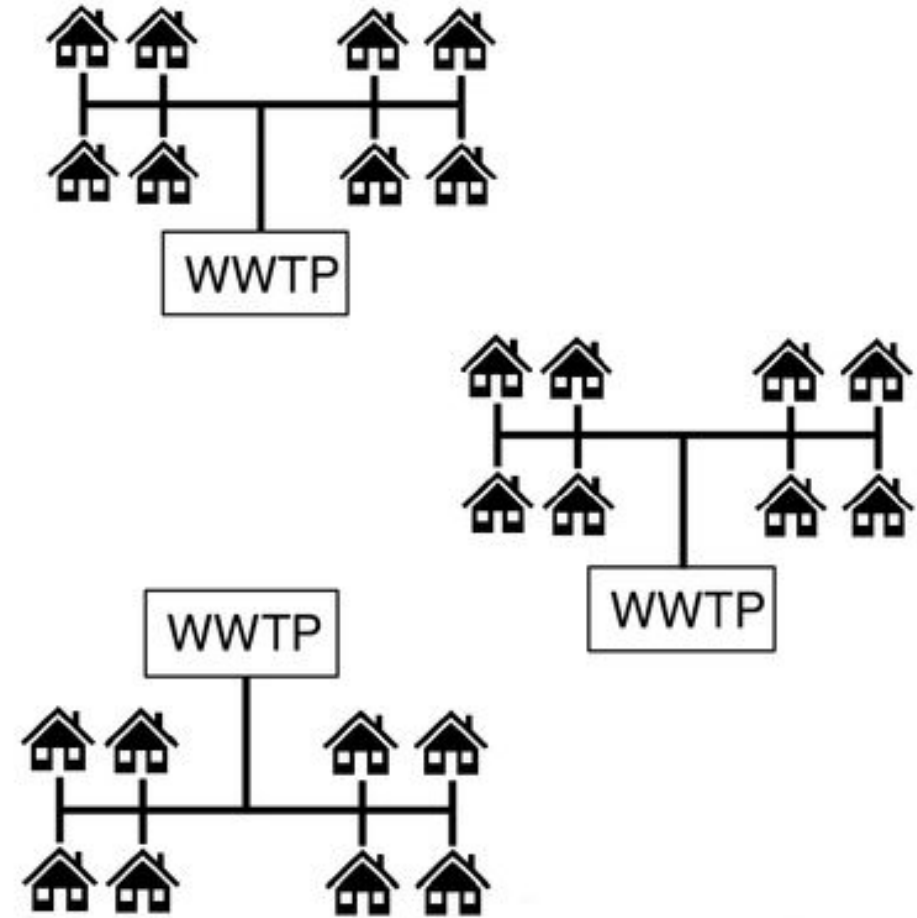
Wastewater/Sewage treatment plant (WWTP/STP)

- An efficient treatment scheme that is capable of **reliably meeting water quality objectives/standards**
- The degree of treatment required varies according to:
 - **Wastewater composition**/characteristics
 - **Associated effluent/water quality requirements** (e.g., environmental discharge, groundwater recharge, irrigation, inland surface water, marine coastal areas, etc.)
 - Specific **reuse** application

Centralized treatment

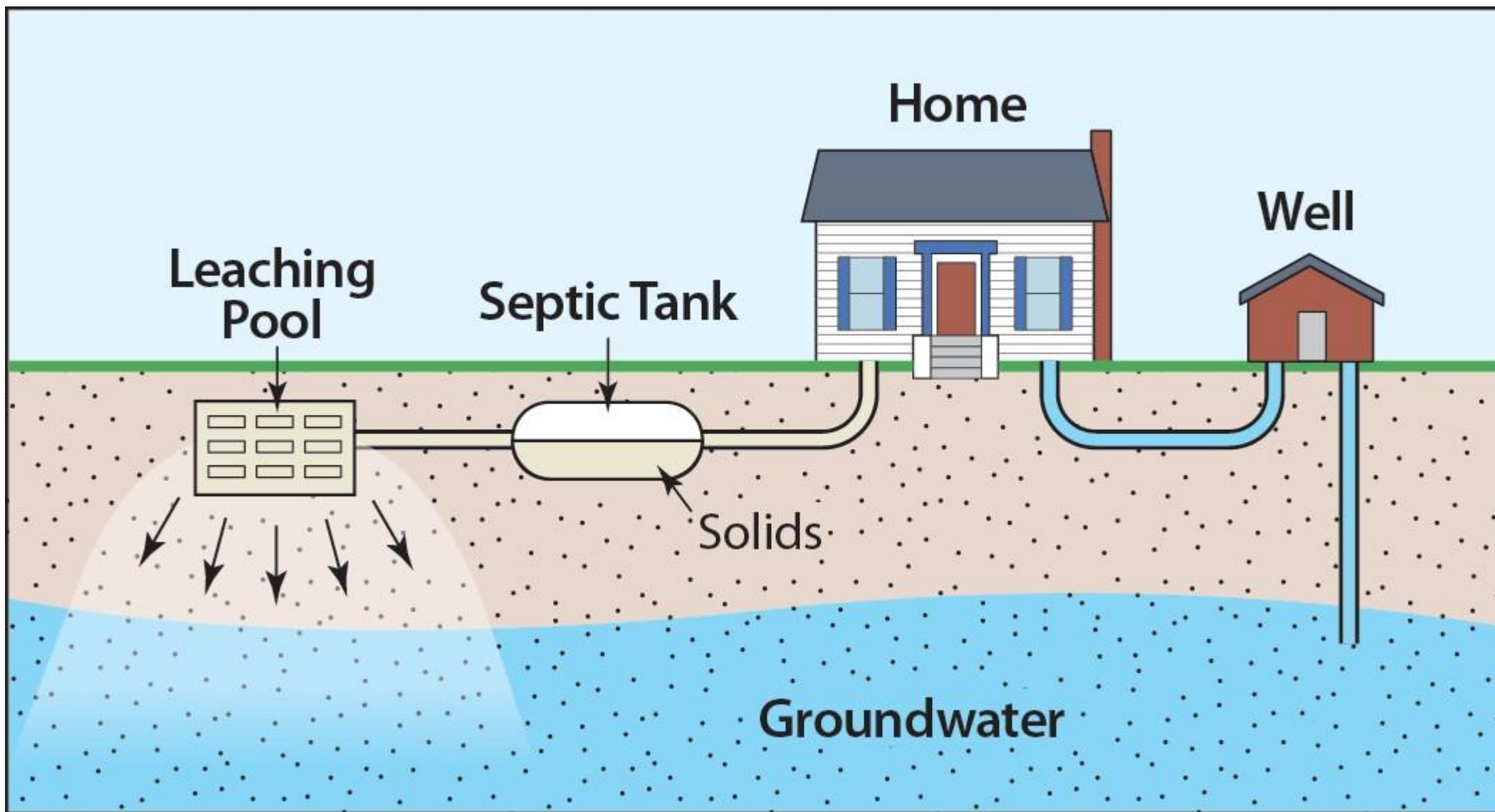


Decentralized treatment

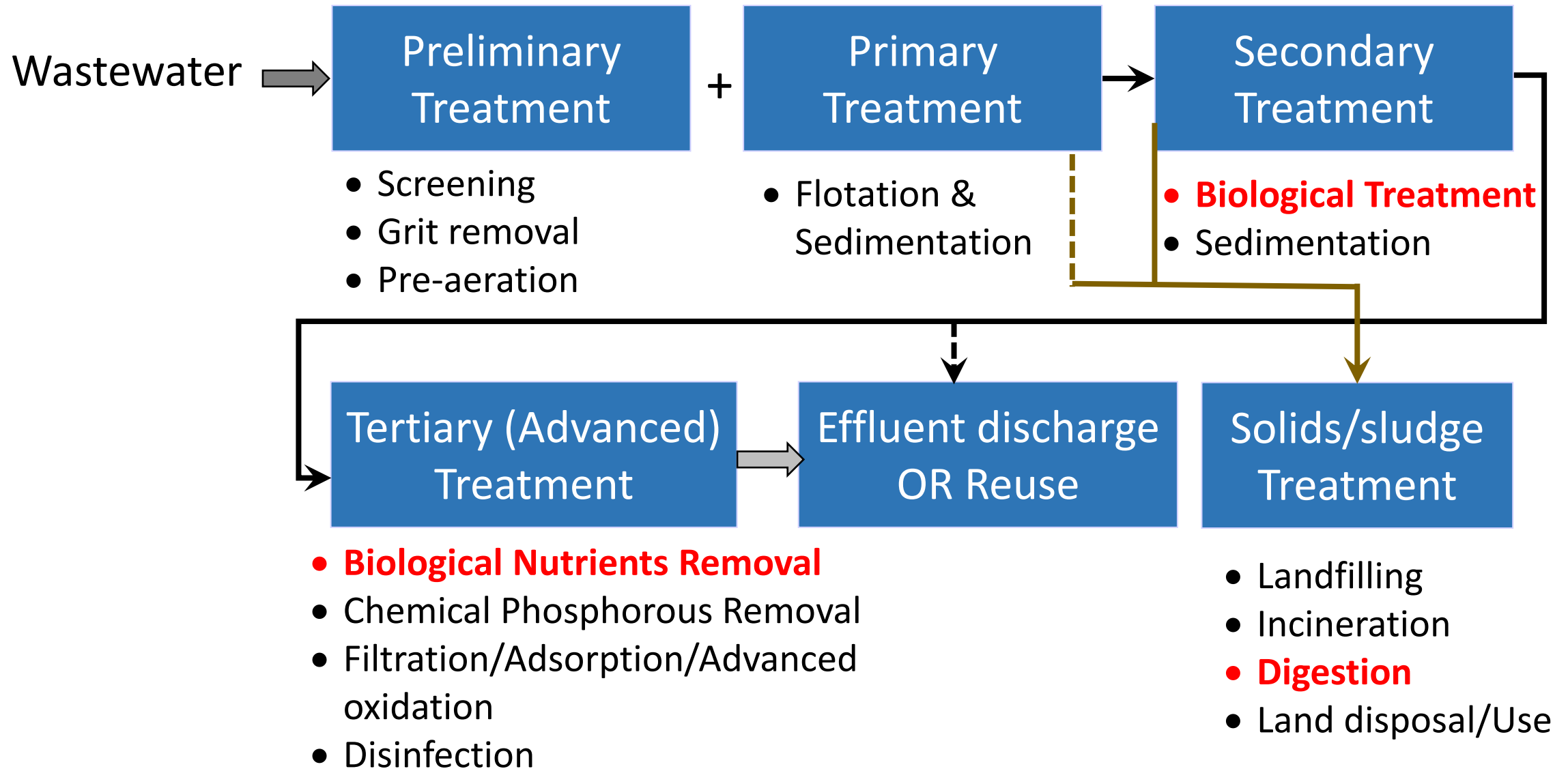


Decentralization is becoming a preferred strategy because it enables right-sized treatment units tailored to specific needs, scalability, upgradability, & resource recovery

On-site wastewater treatment systems (OWTS)



Wastewater treatment – a multistep/process operation



Primary treatment

Sedimentation/Settling

- wastewater flow is slowed down
suspended solids and particulate organic/
inorganic matter settles at the bottom by gravity
- the settled material is referred to as **primary sludge or solids**



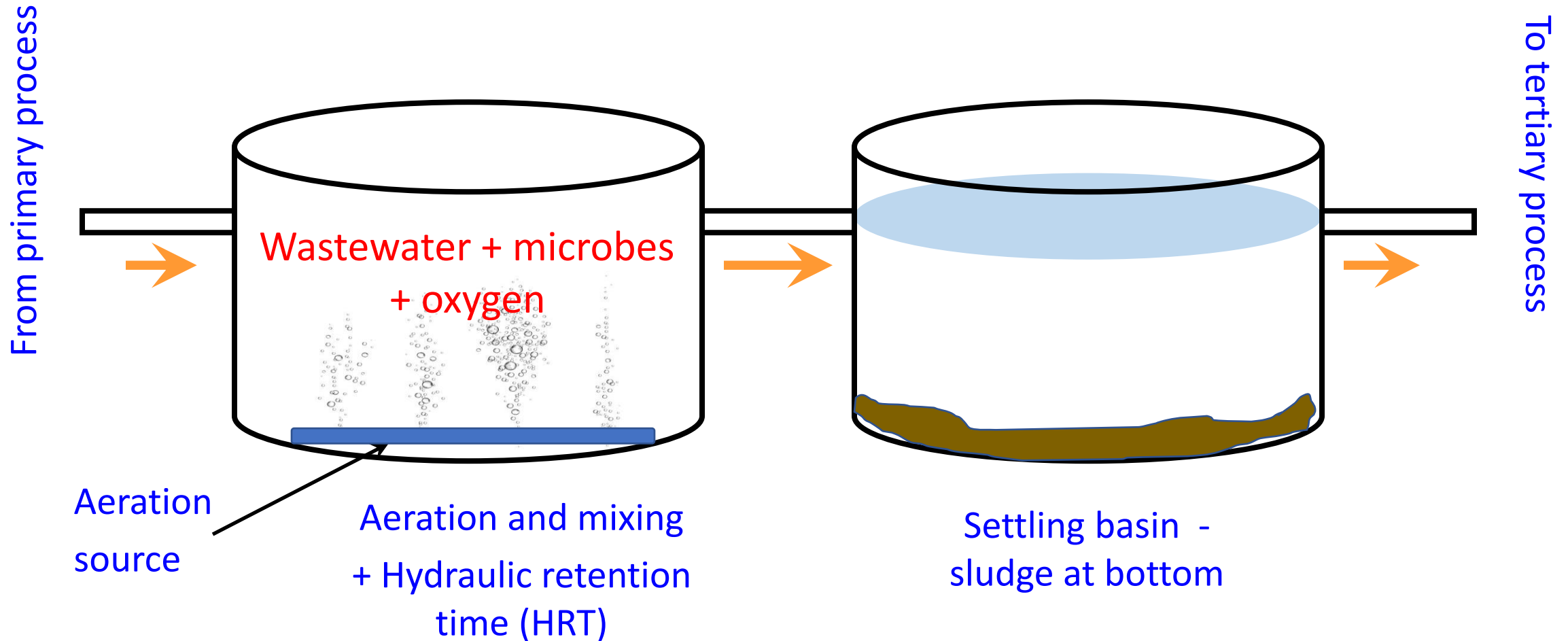
Floatation

- oily and floating materials are removed
- the clarified effluent is referred to as **primary effluent**

Secondary treatment

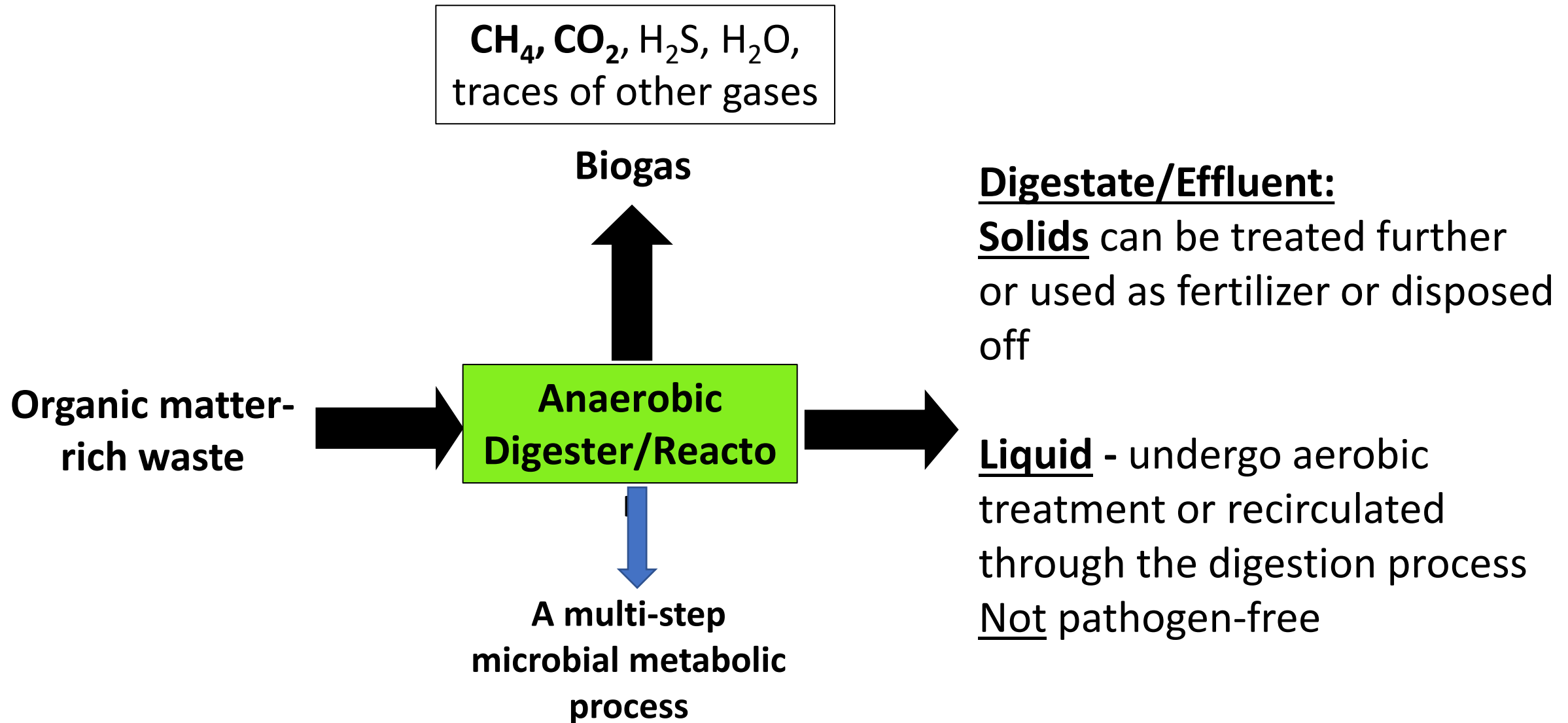
- Removal of organic matter - **Biological treatment**
- Sludge/Biosolids accumulation - **Sedimentation**
- **Biological treatment**
 - Biodegradation of organic matter present in the primary effluent
 - Indigenous microflora (or added/retained)
 - **Aerobic treatment**
 - **Anaerobic treatment**
 - Combination of aerobic and anaerobic processes/Integrated processes

Typical aerobic treatment process



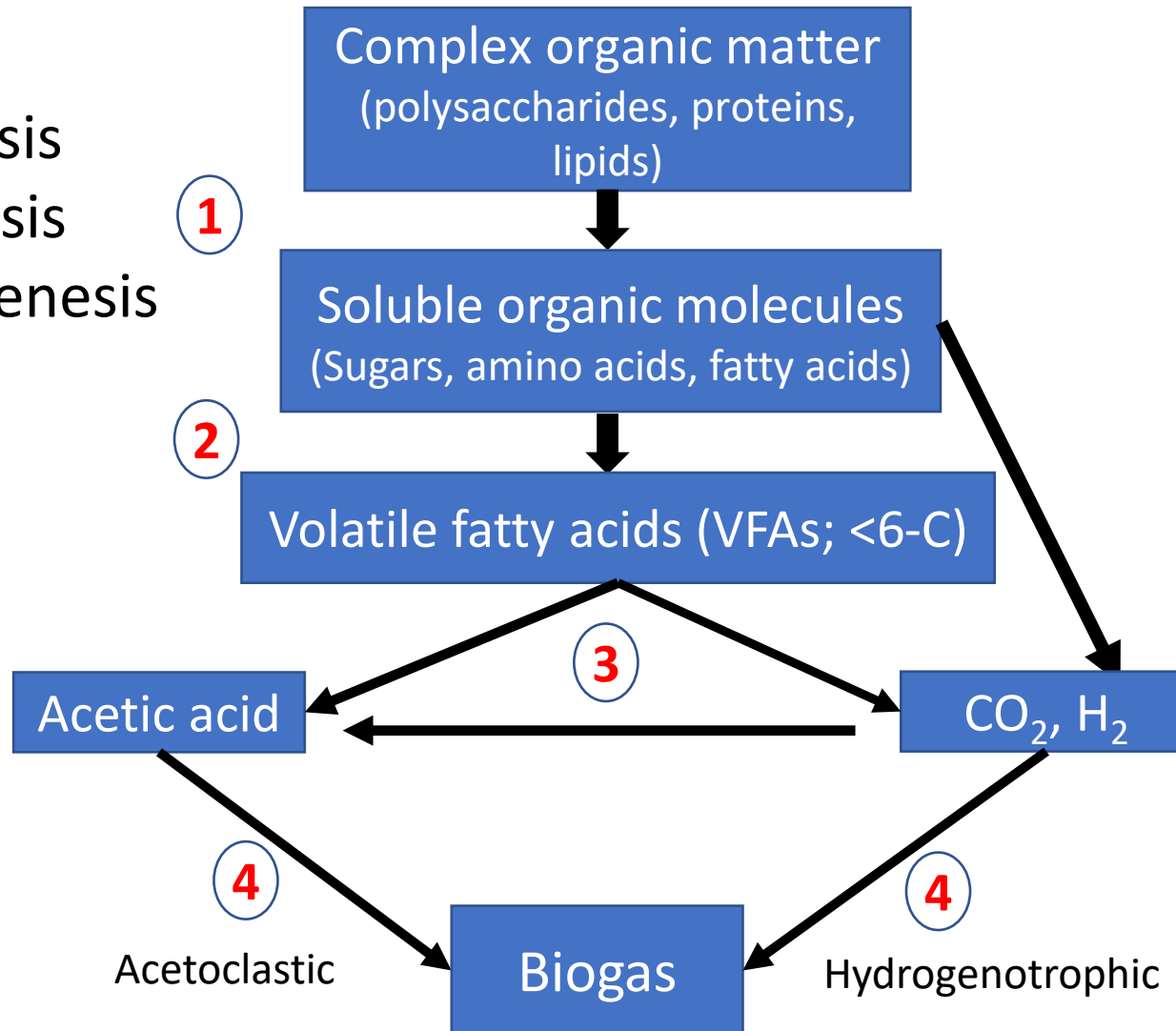
*HRT: Average time length or duration during which (waste)water stays in a reactor

Typical anaerobic digestion/treatment process



Anaerobic digestion (AD) process

1. Hydrolysis
2. Acidogenesis
3. Acetogenesis
4. Methanogenesis

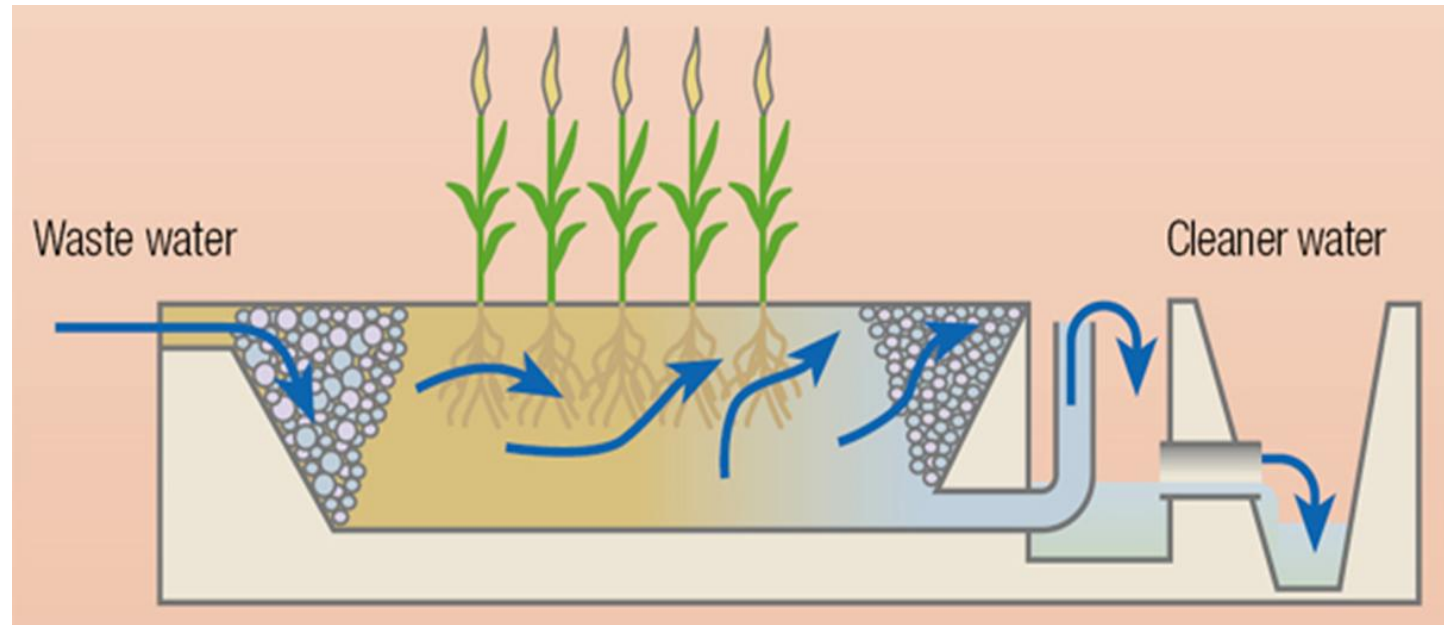


- Removal of organic matter is accomplished by the conversion of organics into CH₄ & CO₂

- AD does not lead to complete conversion - some organic matter remains (digestate)

Nature-based solutions: Constructed Wetlands

- Human-made large gravel and sand-filled basins with vegetation
- Macrophytes (water-loving plants), microorganisms, gravel, and environmental components
- Waste removal mechanisms:
 - Biological
 - microbial degradation
 - plant nutrient uptake
 - Physicochemical
 - adsorption
 - sedimentation
 - filtration



- Passive treatment
 - No electricity
 - No chemicals
- Low-cost construction with local materials
- Low operation and maintenance costs
- Slow treatment – more land requirement
- Environmental factors fluctuate treatment efficiency
- Clogging issues



CRPF, Bhubaneswar

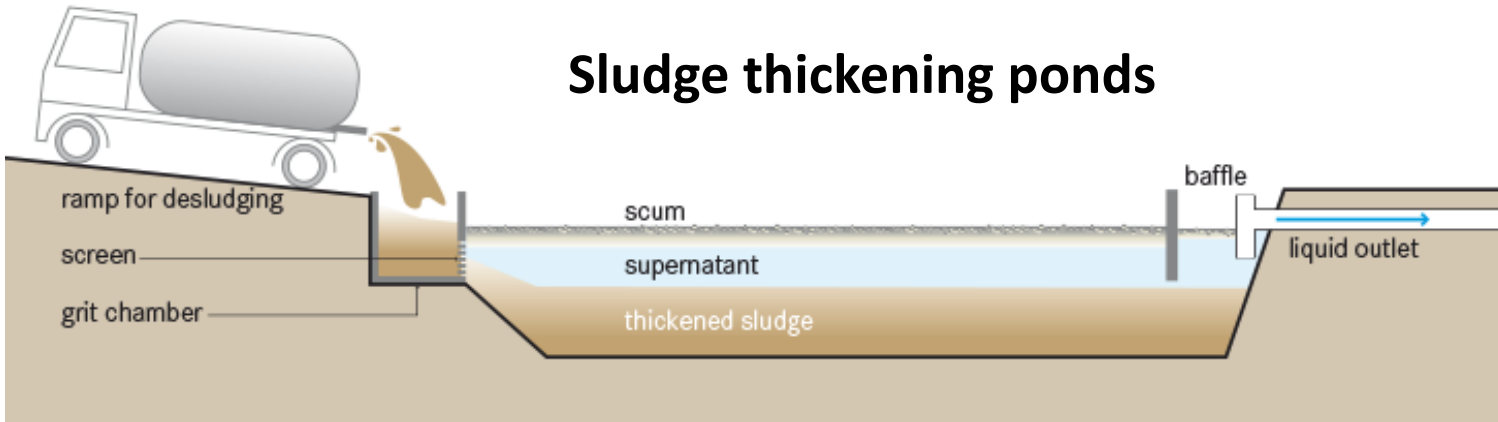
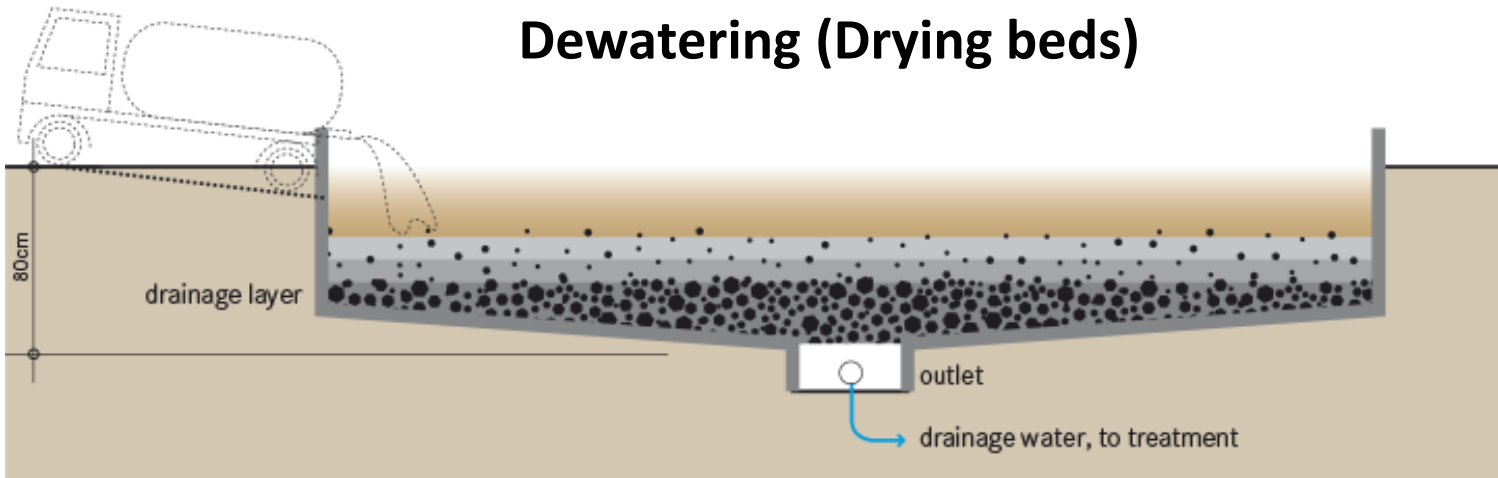


CSIR-IMMT, Bhubaneswar

Tertiary or advanced treatment

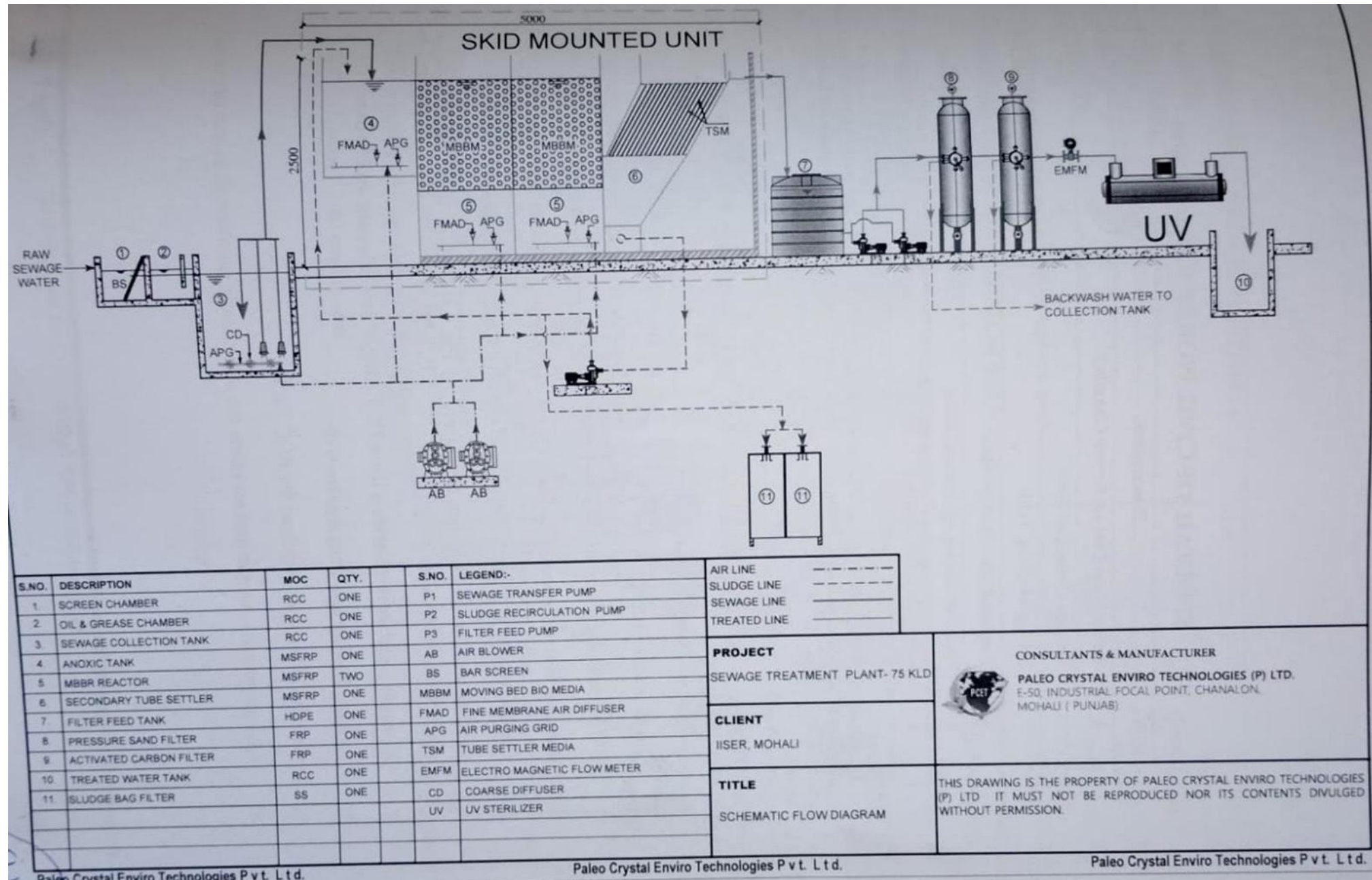
- Applied for removing specific wastewater constituents that cannot be removed by the secondary treatment processes, or obtaining the effluent of a higher quality than normally achieved by the secondary treatment process is desired.
- Unit Processes/Operations:
 - **Biological Nutrient Removal**
 - Physical and chemical processes (e.g., Precipitation, Adsorption, Filtration, Advanced oxidation, etc.)
 - **Sludge treatment**
 - Disinfection (as discussed earlier)

Sludge/Solids management processes



- Incineration (ash and gases production, heat/energy recovery)
- **Digestion**

Representative IISER Mohali decentralized STP (near IWD)

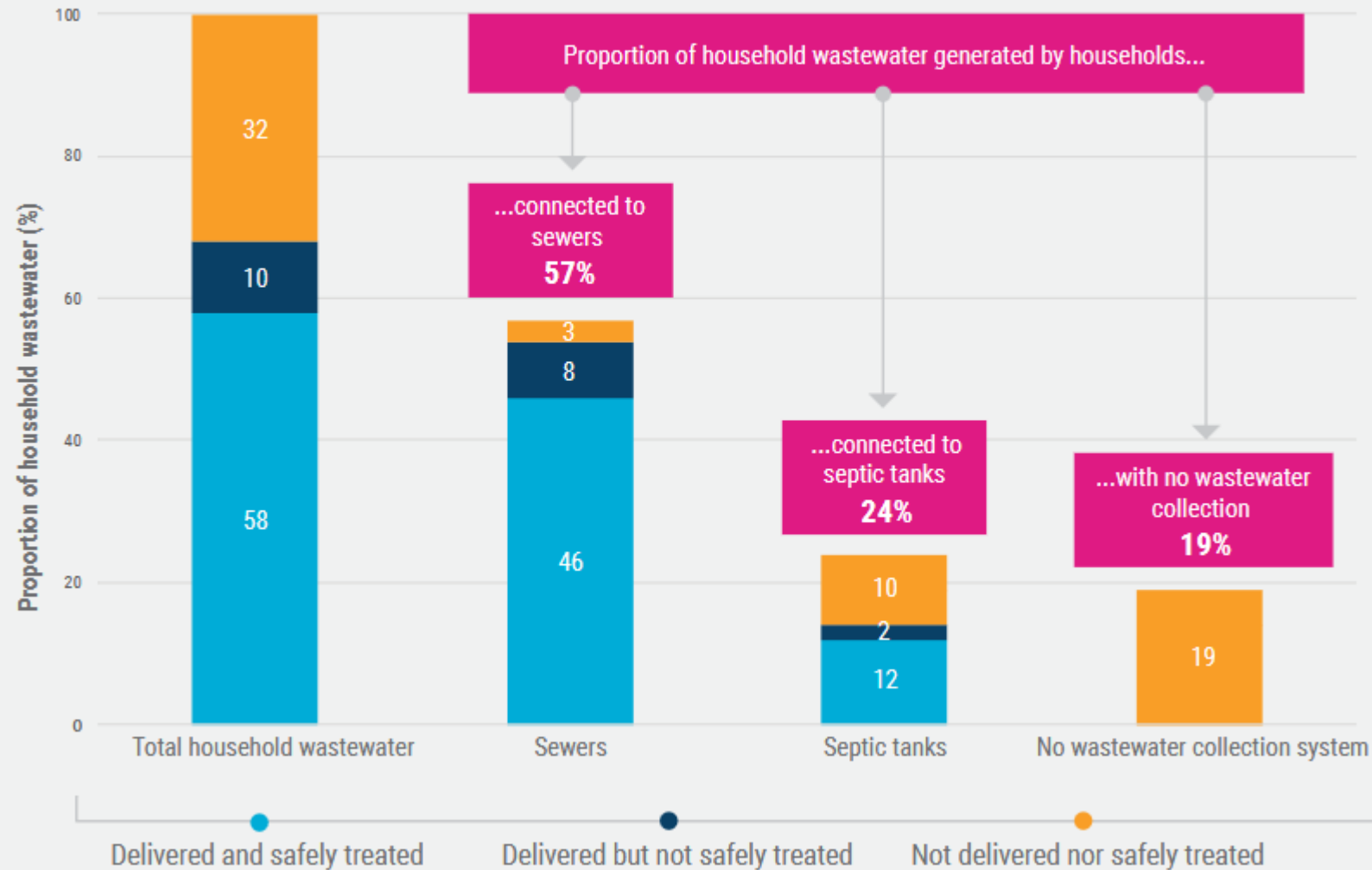


Treatment according to the reuse purpose

Reuse	Treatment
Agriculture	Preserve as much nutrients as possible, kill the pathogens (low-cost technologies)
Domestic (Flushing toilets, gardening etc.)	Moderate to high degree treatment, no pathogens
Industrial use	Higher degree treatment (Advanced)
Groundwater recharge	Needs high-end treatment if the groundwater table is high, otherwise moderate treatment ok; the subsurface soil acts as a filtration unit
Base flow in rivers	Moderate, depends on the carrying capacity of the river
Drinking purpose	Higher degree treatment (Advanced)

Current wastewater management scenario (global)

Most household wastewater flows that are not safely treated are never collected



Source: UN & WHO, 2024

- Intentional Reuse:
 < 15 %
- Limited recovery of resources:
 Energy, Nutrients
 (N & P), Reclaimed water, etc.
- **Future outlook –**
 transitioning
 wastewater treatment
 facilities **into resource
 recovery facilities**

Dedicated EES courses on this topic

- IDC408: Environmental hygiene, sanitation and **waste management**
- EES417: Fundamentals of water and **wastewater treatment**
- EES643: Environmental biotechnology (**focus on Resource Recovery**)
- EES311: Air, **water** and soil analysis lab

Solid Waste Management (SWM)

Source-based broad types of solid wastes

- **Municipal solid waste (MSW):** waste from households, construction and demolition sites, sanitary sites, and streets – **primarily from residential and commercial complexes.**
- **Industrial solid waste:** can be hazardous wastes as it may contain toxic/corrosive/highly inflammable/reactive substances when exposed to certain things, e.g., gases – managed separately
- **Biomedical waste:** Hospital and laboratory waste that includes sharps, soiled waste, disposables, pathological and anatomical waste, cultures, discarded medicines, chemical wastes, etc., usually in the form of disposable syringes, swabs, bandages, body fluids, etc. – managed separately

Material-based MSW categorization

PLASTIC



ORGANIC



GLASS



E-WASTE



PAPER



METAL



Other types:

Source-based,
Hazardous/Non-hazardous,
Recyclables/Non-recyclables,
biodegradable/Non-biodegradable

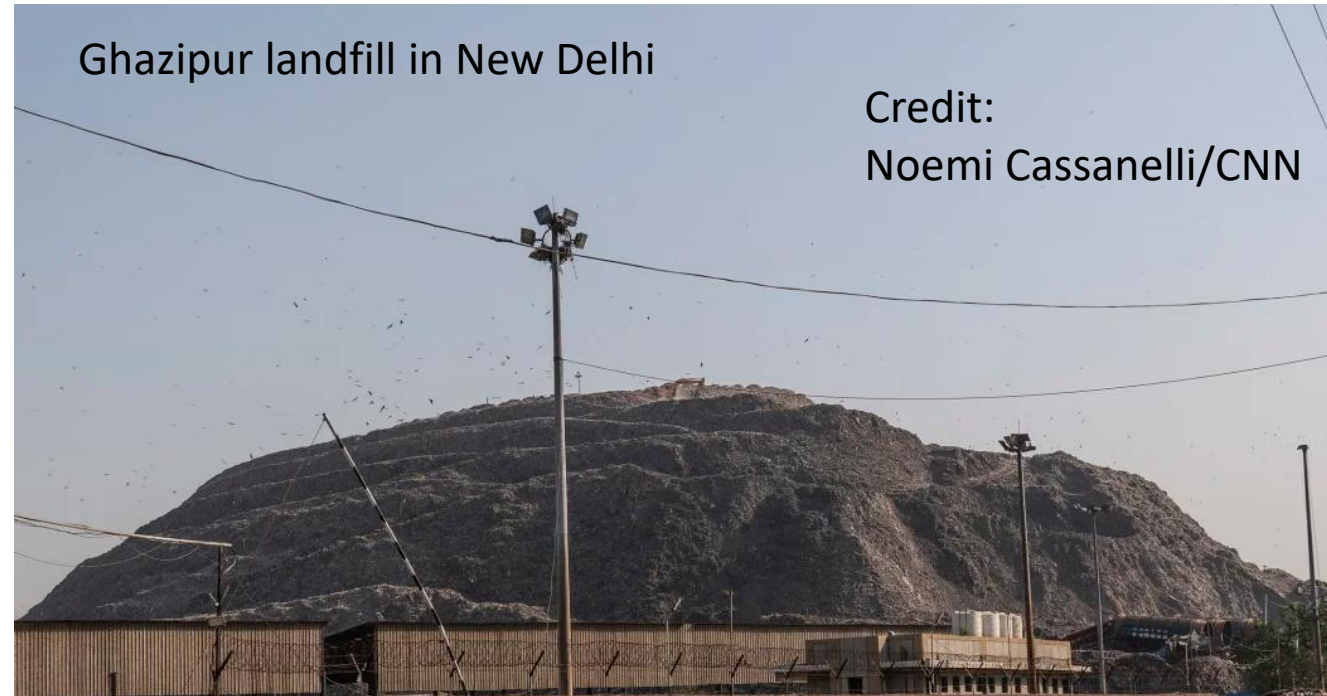
A general SWM scheme includes:

- Collection/segregation at source
- Transportation
- Segregation (at a centralized facility) – Material Recovery Facilities
- Recycling
- **Treatment and Disposal**
- Waste minimization and control

SWM scenario in India (per year)

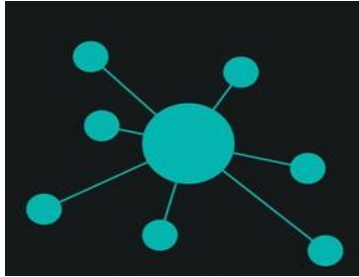


- Around **30% of the total solid waste remains uncollected**
- Collection and transportation - incurs **70% of the total operational cost**
- Most of the collected waste is **landfilled** due to the lack of proper segregation and inadequate infrastructure

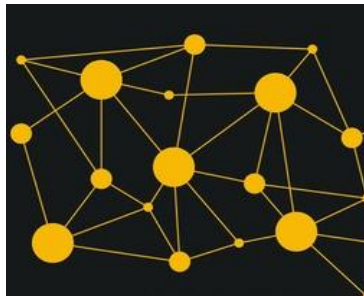


- **About 250 million tons of legacy waste**

SWM management approaches



Centralized



Decentralized

WASTE SEGREGATION CHART

Wet Waste 	Vegetable & Fruit Peel, Food Remains, Expired Food item, Meat, Bone, Egg Shell, Flower, Tea Bag & Coffee Powder, Coconut Shell & Fibre etc.	Do not Put in Plastic Cover They can be composted
Dry Waste 	Plastic, Paper, Wood, Glass, Rubber, etc items not having medical or sanitary residues.	Clean and Store. Give it to Recyclers
Sanitary Waste 	Used Sanitary Napkins, Diaper, Dead Pest, Ear Buds, Dental Floss, Bandage	Do not throw as is. Wrap in Newspaper. Mark it with red cross (X) before disposal
Garden Waste 	Large Quantity of Leaves, Branches, Dried Plants	Handover Separately to BBMP Collection Trucks
Hazardous Household Waste 	Medicines, Pesticides, Old Paints, Hair Colour Mosquito Repellent, Syringes, Cosmetics	DO Not Mix with other Waste. Store Carefully. Hand it over to specific Recyclers
Debris/ Rubbish 	Construction Debris, Demolition Waste, Broken Glass, Broken Furniture	Call BBMP or Other Agencies for Pickup
E-Waste 	Tube lights, CD's, Batteries, Computers, Televisions, Mobile phones, Laptops, Printer Cartridges, Cables	DO Not Mix with other Waste. Store Carefully. Hand it over to specific Recyclers

Source: Swachha, Bengaluru (BBMP)

Integrated solid waste management hierarchy



Source: EPA

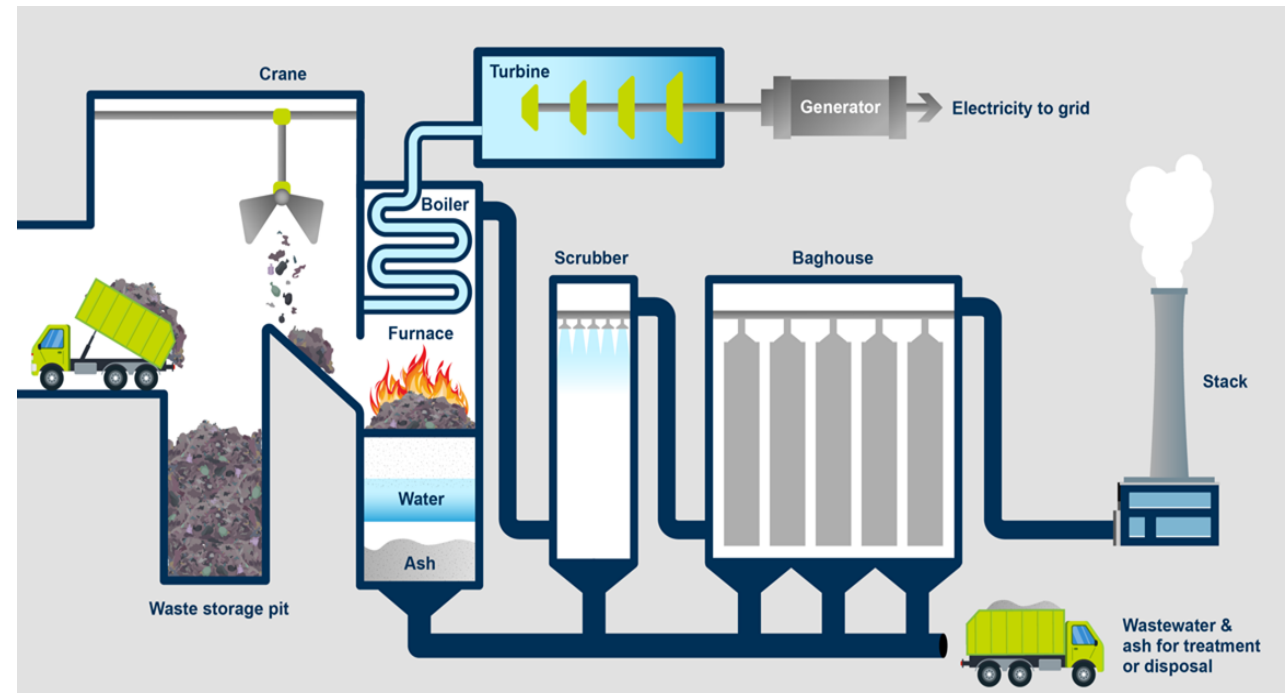
Major solid waste treatment technologies

- **Thermal**
 - Incineration
 - Pyrolysis
 - Gasification
- **Biological**
 - Composting
 - Anaerobic digestion/Bio-methanation (discussed earlier)
- Open **dumps** and Sanitary **landfills**

Incineration

- Combustion of waste in the presence of O_2
- Temp $>850\text{ }^{\circ}\text{C}$
- Products: **heat**, flue gas, and ash (non-combustible materials).
- **Waste to Energy (WtE)**: utilizing the combustion heat through a boiler to generate steam – steam turbine – electricity generator

- Waste reception and handling
- Combustion chamber
- Energy recovery plant
- Emissions clean-up for combustion gases
- Ash and air pollution control residues handling

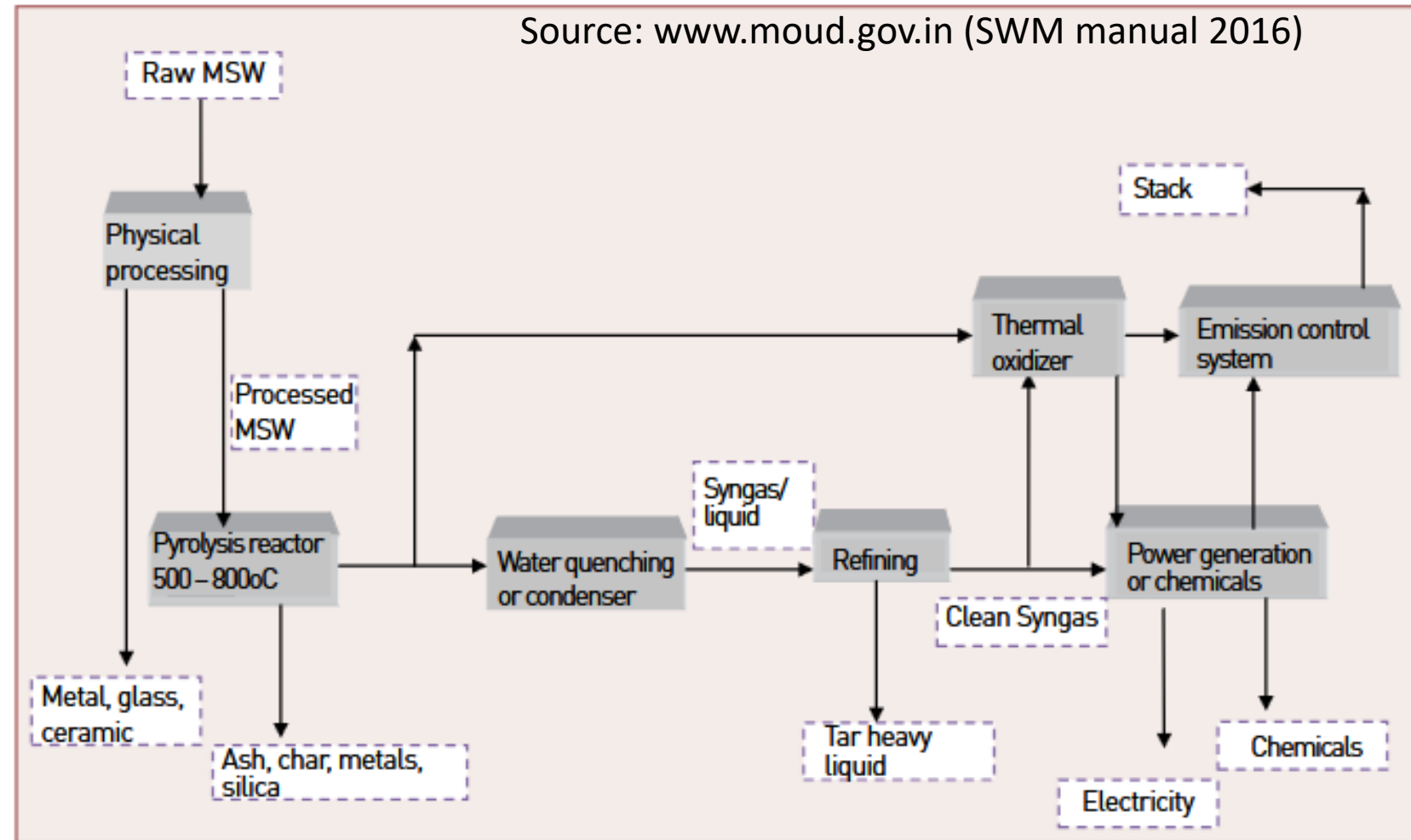


Emissions clean-up for combustion gases: Common approaches

- Ammonia injection into the hot flue gases for control of NO_x emissions
- Lime or sodium bicarbonate injection for controlling SO₂ and HCl emissions
- Activated carbon for capturing heavy metals and other particles
- Filter system for removing fly ash and other solids (lime or bicarbonate and carbon)

Pyrolysis

- Thermal processing is in the complete absence of oxygen
- Temp: 500 - 850 °C
- Products: solid residues/char, **synthesis gas (syngas)**, Liquid



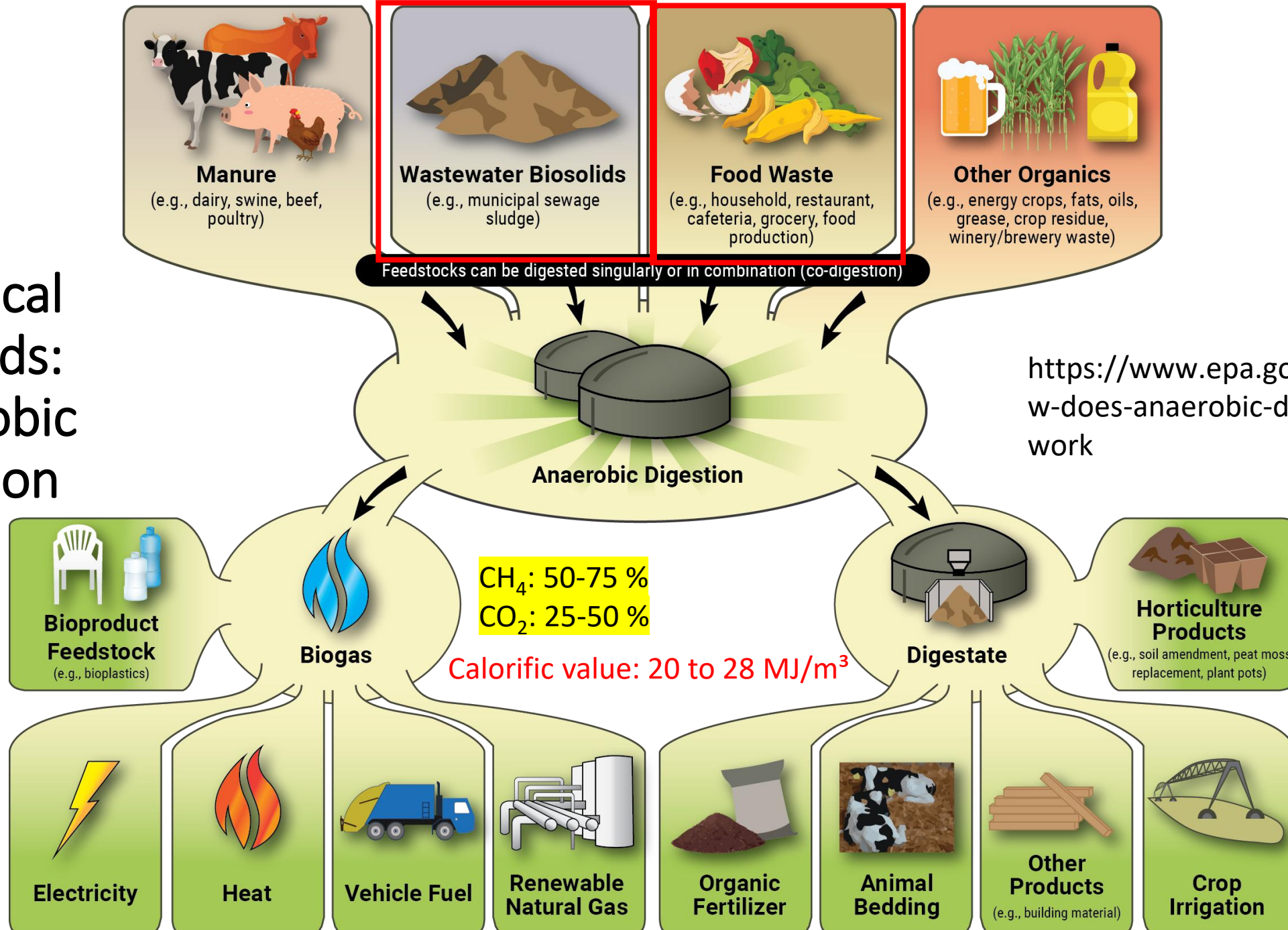
Syngas - a mixture of gases, mainly CO and H₂
(calorific value 4 – 12 MJ/m³)

- **Bio-oil (Pyrolysis Oil/Bio-crude):** dark brown liquid having a similar composition to biomass; upgradation to synthetic fuel
- **Biochar:** various applications, such as soil amendment, as a sorbent in water filtration, carbon storage, etc.
- **Syngas:** It has a high calorific value, so it can be used as a fuel to generate electricity or steam and also as a basic/platform chemical in the chemical industry.

Gasification

- Partial oxidation/combustion of carbonaceous waste
- Temp: $>650\text{ }^{\circ}\text{C}$ and limited O_2 supply
- Products – **Syngas** (CO , H_2 , CO_2 and CH_4) and solid residue (char/ash)

Biological methods: Anaerobic Digestion



Biological methods: Composting

- The organic matter is decomposed microbially under controlled conditions – reduction in waste quantity
- The stabilized product is called compost or humus – it resembles potting soil in texture and odor and can be used as a low-grade manure or soil conditioner.
- **Steps:** sorting, size reduction/shredding, mixing, and digestion of waste.
- **Methods:** Windrow composting, Aerated static pile composting, In-vessel/Mechanical composting, Vermicomposting, etc.

Windrows composting:

- Rows of long piles - turned or mixed every few days to provide air to enable microbial organics degradation
- 5-8 weeks for complete digestion, slow process
- Temperatures in an active compost pile reach about 65 °C, killing non-spore-forming pathogenic microbes present in the waste
- Open windrow composting is weather-dependent and requires relatively large land areas



In-vessel composting/Enclosed mechanical composting:

- One or more closed tanks equipped with a shredder and rotating blades that mix and aerate the shredded waste under controlled temperature, moisture, etc. conditions.
- Less time for complete waste digestion
- Can reduce land requirements by about 85 %
- High CAPEX and energy use



Source: Eco Paryavaran

- ✓ Onsite treatment; Less manpower
- ✓ No effect of weather; Low odor
- ✓ Faster decomposition & more consistent product quality
- ✓ Process air & leachate can be easily managed

Open Landfills or Dumps

- Open areas expose solid waste to natural elements, stray animals, and birds.
- Absence of any kind of monitoring and leachate collection system - contamination of both land and water resources.
- Release of GHGs and other gases, e.g., CH_4
- >3000 open dumps (CSE India, 2021)

Mumbai's Deonar dump is 18 storeys high (120 ft)



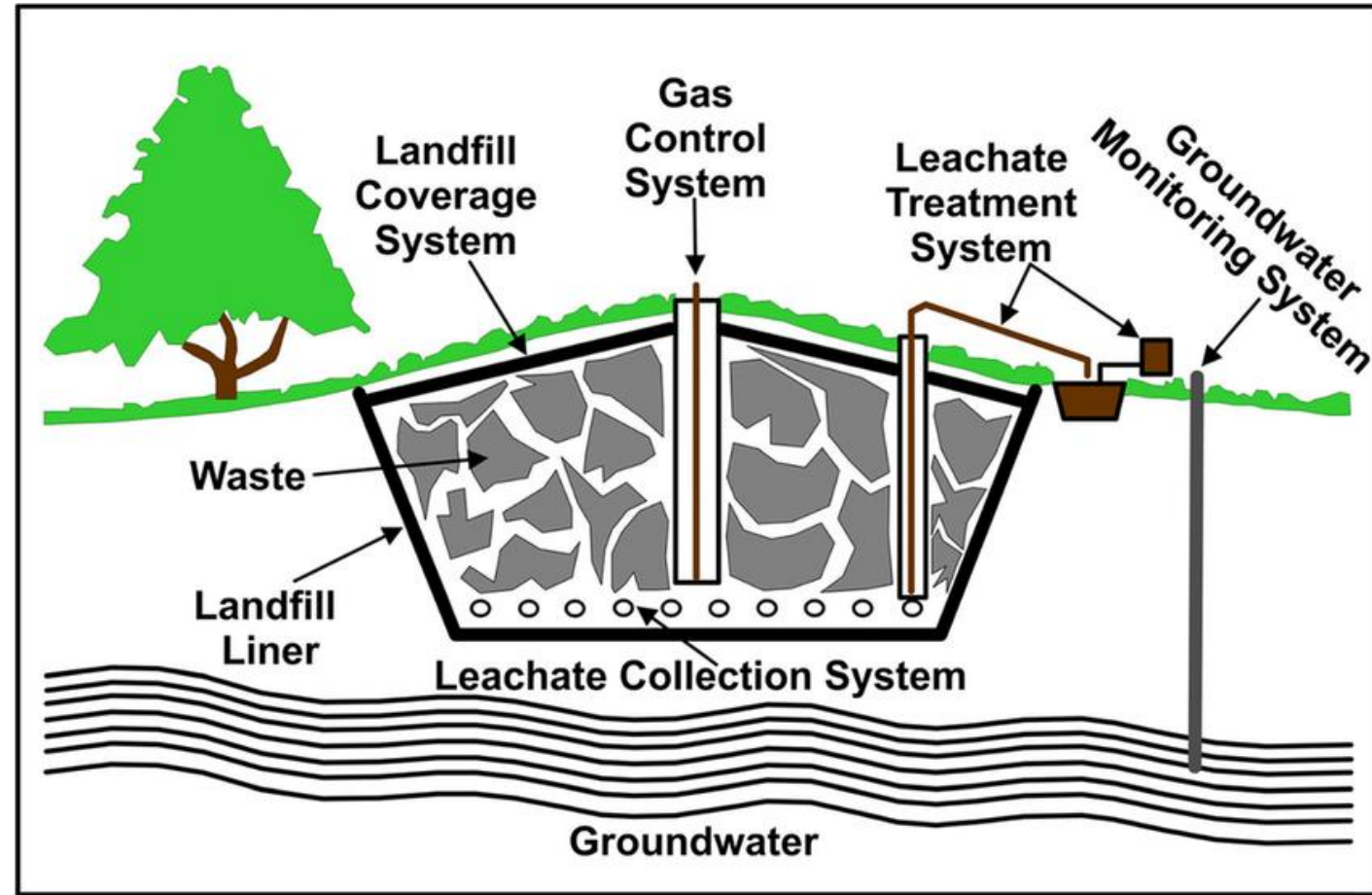


Bhalswa landfill
site (62 meters -
203 feet high) on
April 28, 2022, in
New Delhi

(Sanchit Khanna,
Hindustan Times/Getty
Images)

Sanitary Landfills

- Carefully selected, designed, constructed, and operated to protect the environment and public health.
- Equipped with leachate and gas collection and management systems
- Impermeable bottom liners and top cover to protect groundwater contamination and surface runoff, respectively.



Arshad et al. Environmental Science and Pollution Research 2023, 30

- Different design variations and types

Exemplary models for SWM in India

- Allappuzha
- Indore
- Mysuru

Decentralized: Alappuzha

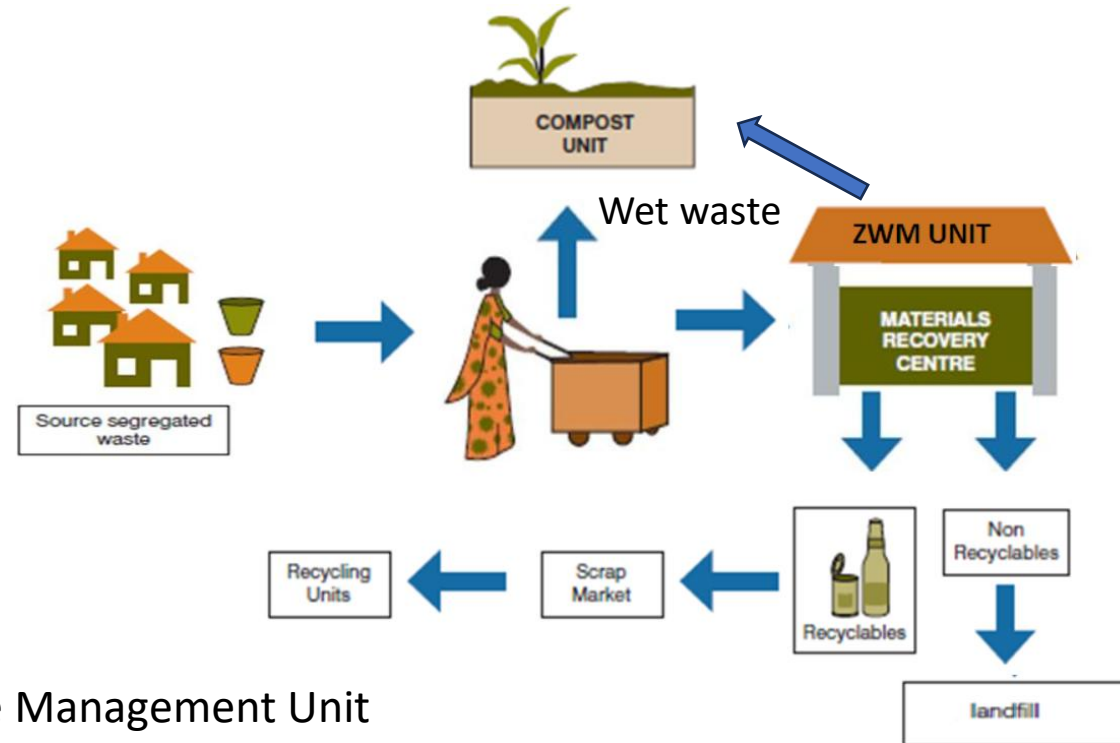
- Alappuzha in Kerala has a population of ~0.174 million and produces 58 MT of solid waste/day
- It has adopted decentralized waste management and is pushing for 100 % segregation in all 23 wards.
- Scheme Nirmala Bhavanam Nirmala Nagaram (Clean Homes Clean City) since November 2012: 80 % of households have biogas plants and a decentralized composting system
- The non-biodegradable waste is sent to certified recyclers.



<https://www.downtoearth.org.in/news/waste/alappuzha-gets-recognized-by-unep-for-its-solid-waste-management-practices-59258>

Mysuru

- The city generates around 400 MT of total solid waste/day, of which 60 % is wet waste, and the rest is dry waste.
- A model involves door-to-door collection of segregated waste, followed by segregation at a central facility, recycling/reuse, and wealth from waste approaches.



ZWM: Zero Waste Management Unit

Indore (Madhya Pradesh)

- It produces >1,115 MT of solid waste/day, and every last bit of it is collected from the source
- Segregation at source (wet waste, dry waste, and hazardous waste)
- Modern and efficient infrastructure
- Waste Management: Gasification, Bio-methanation, Incineration, Composting, Recycling, Sanitary Landfills, etc.
- Public awareness and participation
- Strict monitoring and enforcement



<https://smartcityindore.org/solid-waste/>



Key challenges in SWM in India

- Conflicting data
- Failure to establish efficient systems and technologies, plus a lack of awareness
- Local authorities lack adequate funding and infrastructure - unable to adopt innovative and appropriate technologies
- WtE is a widely used technology, but it faces several problems - unsegregated waste and seasonal variation in waste composition, operational and design issues - most WtE plants do not function effectively
- Shortage of PPE/safety gears for staff - undermines the safety of SWM employees

Key Government Initiatives and Policies

- **Solid Waste Management Rules, 2026:** Emphasize circular economy, digital tracking, and 'polluter pays' principles.
- **Swachh Bharat Mission (SBM) Urban 2.0 & Grameen II:** garbage-free cities and sustainable waste management in villages.
- **GOBAR-Dhan Scheme:** cattle dung and organic farm waste into biogas and compost.
- **Plastic Waste Management Rules:** Regulation of plastic waste to encourage recycling and reduce pollution.
- **Waste-to-Energy Program:** Supported by the Ministry of New and Renewable Energy for developing waste-based power plants and Biomethanation projects